STANDARD 2

Stage 6

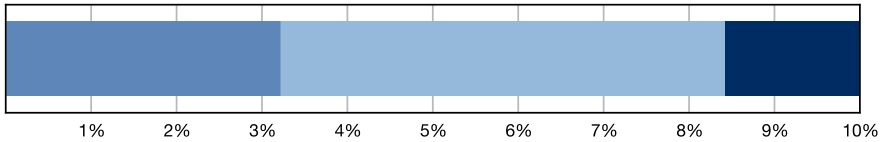
Algebra (Std 2), A4 Types of Relationships (Y12)

Simultaneous Equations and Applications (Std 2)

Teacher: Hailey Summer

Exam Equivalent Time: 96 minutes *(based on allocation of 1.5 minutes per mark)*

STD2 Exam Contribution History
A4 - Types of Relationships



- Simult. Equations and Applications
- Non-Linear: Exponential/Quadratics
- Non-Linear: Inverse and Other Equations

***SmarterMaths analytics based on the average contribution in new syllabus exams (since 2019).**

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-A4 Types of Relationships* has contributed a very substantial 10.0% per paper since the new syllabus was introduced in 2019.
- This analysis looks at *Simultaneous Equations and Applications (3.2%).*

ANALYSIS - What to Expect and Common pitfalls

- *Simultaneous Equations and Applications (3.2%)* is a critical revision area with questions worth 4-5 marks appearing in three of the last four HSC exams (absent in 2022).
- The high mark allocations and low mean marks over this period make it a key area where students can outperform.
- Significant syllabus changes have occurred in this topic area - most notably that students can be tested on solving simultaneous equations graphically, but not arithmetically.
- Cost and revenue graphing is an important theme which has attracted high mark allocations in *2019 Std2 36* and *2018 Std2 27d* - both important revision questions.
- Students must review *2021 Std2 34* which produced a mean mark of just 29% and represents the upper difficulty level of this topic area.
- Past HSC questions have required students to sketch linear equations on 5 separate occasions since 2014 - a key skill to incorporate into any revision.

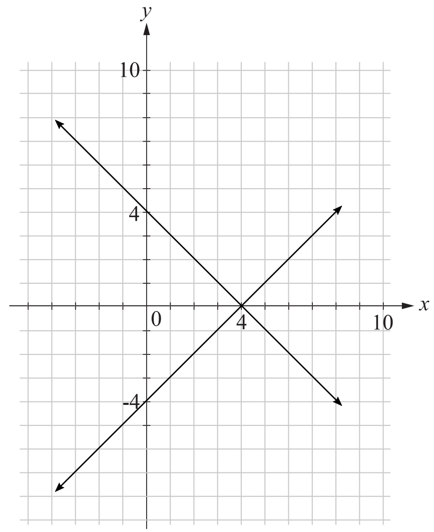
Questions

1. Algebra, STD2 A4 SM-Bank 6 MC

A computer application was used to draw the graphs of the equations

$x - y = 4$ and $x + y = 4$

Part of the screen is shown.



What is the solution when the equations are solved simultaneously?

A. $x = 4, y = 4$

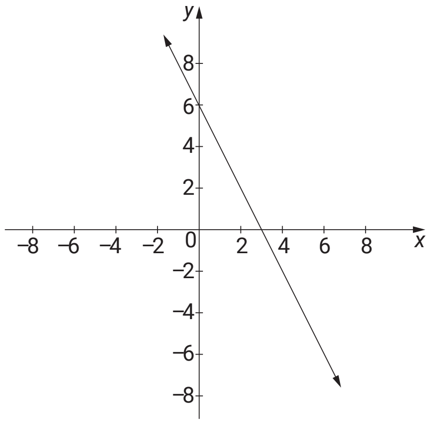
B. $x = 4, y = 0$

C. $x = 0, y = 4$

D. $x = 0, y = -4$

2. Algebra, STD2 A4 2017 HSC 17 MC

The graph of the line with equation $y = 6 - 2x$ is shown.



When the graph of the line with equation $y = x + 3$ is also drawn on this number plane, what will be the point of intersection of the two lines?

A. $(0, 6)$

B. $(1, 4)$

C. $(2, 2)$

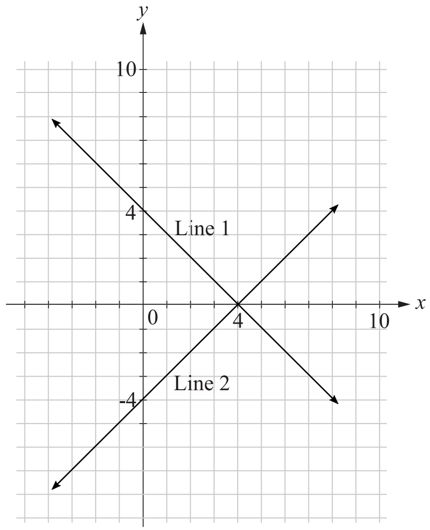
D. $(3, 0)$

3. Algebra, STD2 A4 SM-Bank 7 MC

A computer application was used to draw the graphs of the equations

$x + y = 4$ and $x - y = 4$

Part of the screen is shown.



Which row of the table correctly matches the equations with the lines drawn and identifies the solution when the equations are solved simultaneously?

	$x + y = 4$	$x - y = 4$	Solution
A.	Line 1	Line 2	$x = 4, y = 0$
B.	Line 1	Line 2	$x = 4, y = 4$
C.	Line 2	Line 1	$x = 4, y = 0$
D.	Line 2	Line 1	$x = 4, y = 4$

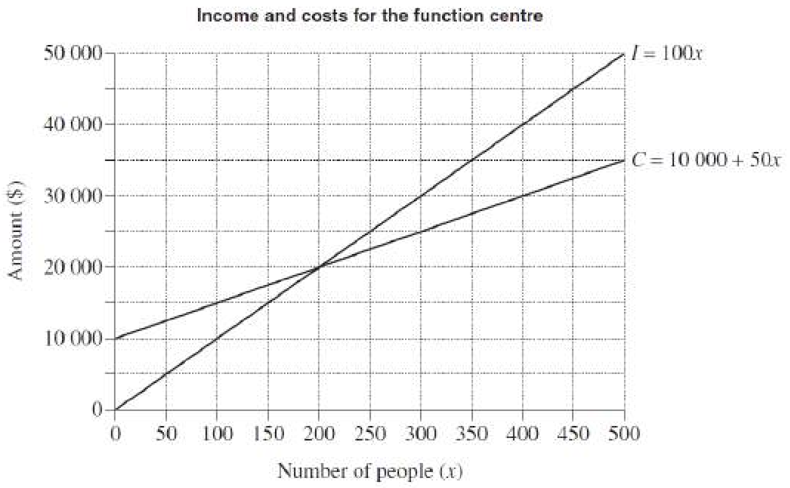
4. Algebra, STD2 A4 2011 HSC 20 MC

A function centre hosts events for up to 500 people. The cost C, in dollars, for the centre to host an event, where x people attend, is given by:

$$C = 10\ 000 + 50x$$

The centre charges $100 per person. Its income I, in dollars, is given by:

$$I = 100x$$



How much greater is the income of the function centre when 500 people attend an event, than its income at the breakeven point?

A. $\$15\ 000$

B. $\$20\ 000$

C. $\$30\ 000$

D. $\$40\ 000$

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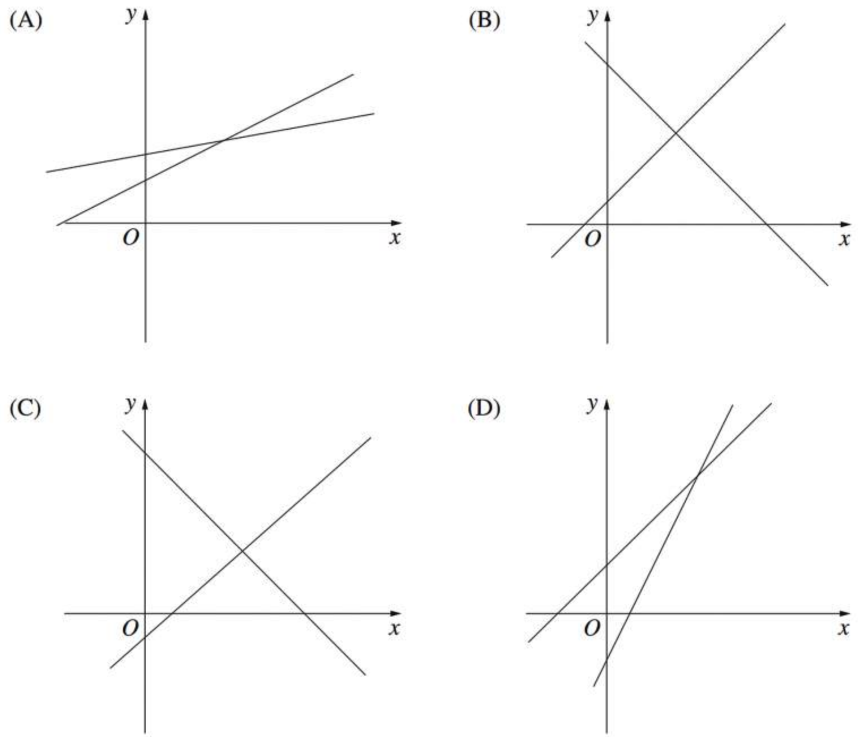
5. Algebra, STD2 A4 2004 HSC 16 MC

George drew a correct diagram that gave the solution to the simultaneous equations

$y = 2x - 5$ and $y = x + 6$.

Which diagram did he draw?



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6. Algebra, STD2 A4 2024 HSC 14 MC

Year 11 is making pizzas to raise money for a charity.

The cost (C) and revenue (R) in dollars, when x pizzas are sold are represented by the equations

$$C = 2.5x + 6$$
$$R = 8x.$$

Enough pizzas are sold so that a profit is made.

By how much does the profit increase for each additional pizza sold?

A. $2.50

B. $5.50

C. $6.00

D. $8.00

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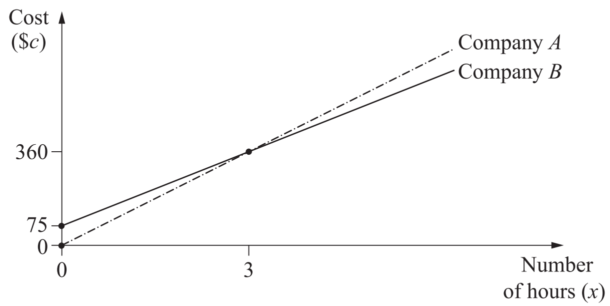
7. Algebra, STD2 A4 2018 HSC 27d

The graph displays the cost ($\$c$) charged by two companies for the hire of a minibus for x hours.



Both companies charge $360 for the hire of a minibus for 3 hours.

i. What is the hourly rate charged by Company *A*? ***(1 mark)***

ii. Company B charges an initial booking fee of $75.

Write a formula, in the form of $c = mx + b$, for the cost of hiring a minibus from Company *B* for x hours. ***(2 marks)***

iii. A minibus is hired for 5 hours from Company *B*.

Calculate how much cheaper this is than hiring from Company *A*. ***(2 marks)***

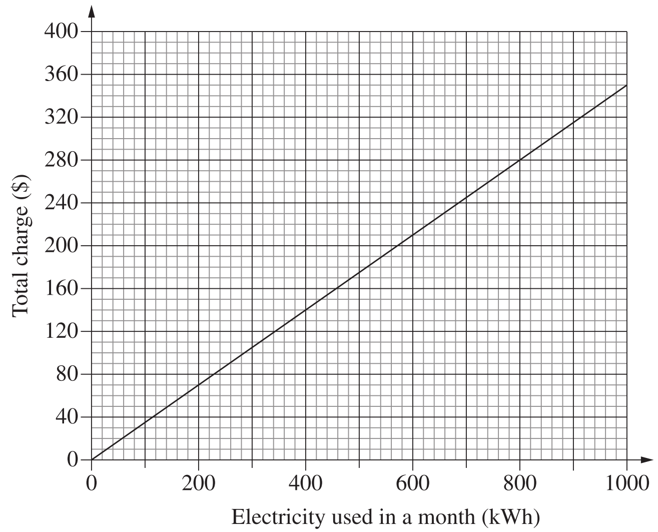
8. Algebra, STD2 A4 2023 HSC 21

Electricity provider A charges 25 cents per kilowatt hour (kWh) for electricity, plus a fixed monthly charge of $40.

a. Complete the table showing Provider A's monthly charges for different levels of electricity usage. ***(1 mark)***

Electricity used in a month (kWh)	0	400	1000
Monthly charge ($)	40		290

Provider B charges 35 cents per kWh, with no fixed monthly charge. The graph shows how Provider B's charges vary with the amount of electricity used in a month.



b. On the grid above, graph Provider A's charges from the table in part (a). ***(1 mark)***

c. Use the two graphs to determine the number of kilowatt hours per month for which Provider A and Provider B charge the same amount. ***(1 mark)***

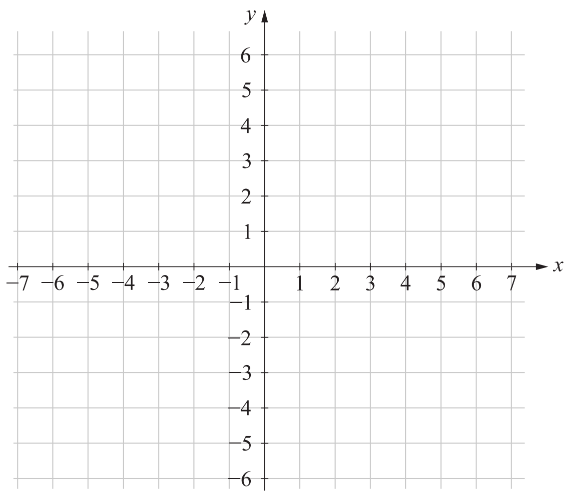
d. A customer uses an average of 800 kWh per month.

Which provider, A or B, would be the cheaper option and by how much? ***(2 marks)***

9. Algebra, STD2 A4 SM-Bank 5

$$y = x + 5$$
$$y + 2x = 2$$

Draw these two linear graphs on the number plane below and determine their intersection. ***(3 marks)***

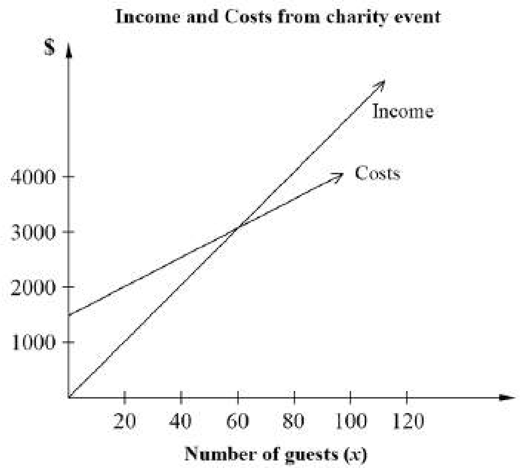


10. Algebra, STD2 A4 SM-Bank 27

Fiona and John are planning to hold a fund-raising event for cancer research. They can hire a function room for $650 and a band for $850. Drinks will cost them $25 per person.

i. Write a formula for the cost ($C) of holding the charity event for x people. ***(1 mark)***

ii. The graph below shows the planned income and costs if they charge $50 per ticket. Estimate the number of guests they need to break even. ***(1 mark)***



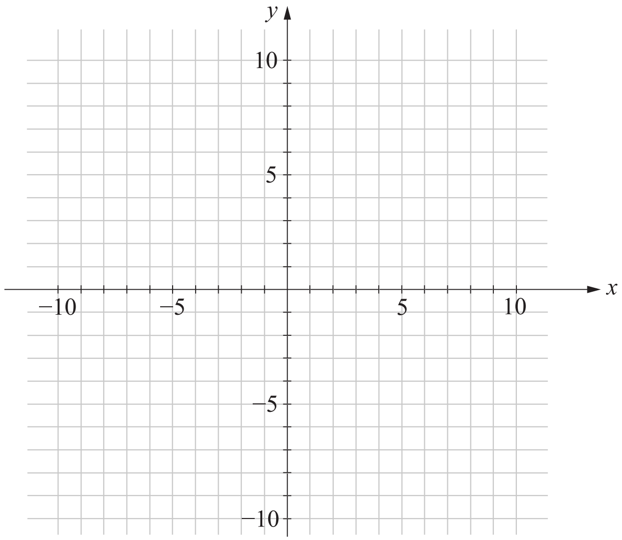
iii. How much profit will Fiona and John make if 80 people attend their event? ***(1 mark)***

11. Algebra, STD2 A4 2014 HSC 26d

Draw each graph on the grid below and hence solve the simultaneous equations. ***(3 marks)***

$y = 2x + 1$

$x - 2y - 4 = 0$ ***(3 marks)***



12. Algebra, STD2 A4 SM-Bank 6

A student was asked to solve the following simultaneous equations.

$y = 2x - 8$

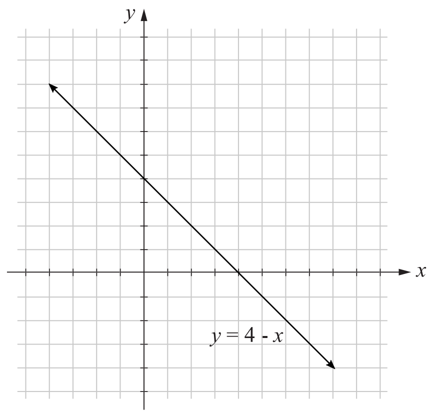
$x - 4y + 3 = 0$

After graphing the equations, the student found the point of intersection to be $(5, 2)$?

Is the student correct? Support your answer with calculations. ***(2 marks)***

13. Algebra, STD2 A4 SM-Bank 7

The graph of the line $y = 4 - x$ is shown.



By graphing $y = 2x + 1$ on the grid provided, find the point of intersection of $y = 4 - x$ and $y = 2x + 1$. ***(3 marks)***

14. Algebra, STD2 A4 EQ-Bank 8

Two friends, Sequoia and Raven, sold organic chapsticks at the the local market.

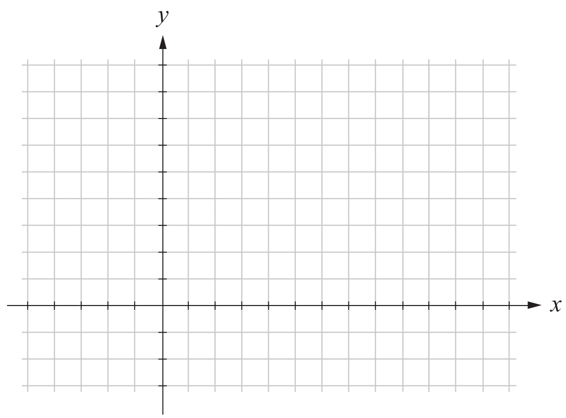
Sequoia sold her chapsticks for \$4 and Raven sold hers for \$3 each. In the first hour, their total combined sales were \$20.

If Sequoia sold x chapsticks and Raven sold y chapsticks, then the following equation can be formed:

$4x + 3y = 20$

In the first hour, the friends sold a total of 6 chapsticks between them.

Find the number of chapsticks each of the friends sold during this time by forming a second equation and solving the simultaneous equations graphically. ***(5 marks)***



15. Algebra, STD2 A4 SM-Bank 3

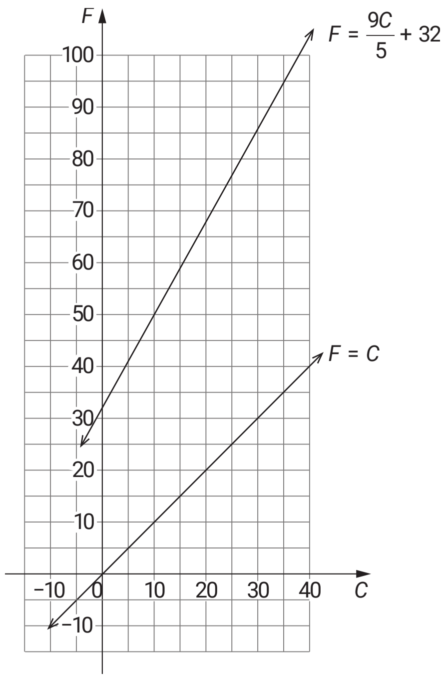
Temperature can be measured in degrees Celsius (C) or degrees Fahrenheit (F).

The two temperature scales are related by the equation $F = \dfrac{9C}{5} + 32$.

i. Calculate the temperature in degrees Fahrenheit when it is −20 degrees Celsius. ***(1 mark)***

ii. The following two graphs are drawn on the axes below:

$$F = \frac{9C}{5} + 32 \text{ and } F = C$$

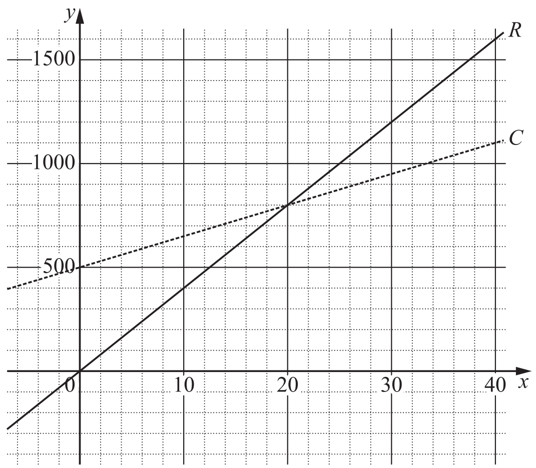


Explain what happens at the point where the two graphs intersect. ***(1 mark)***

16. Algebra, STD2 A4 2019 HSC 36

A small business makes and sells bird houses.

Technology was used to draw straight-line graphs to represent the cost of making the bird houses (C) and the revenue from selling bird houses (R). The x-axis displays the number of bird houses and the y-axis displays the cost/revenue in dollars.



a. How many bird houses need to sold to break even? ***(1 mark)***

b. By first forming equations for cost (C) and revenue (R), determine how many bird houses need to be sold to earn a profit of \$1900. ***(3 marks)***

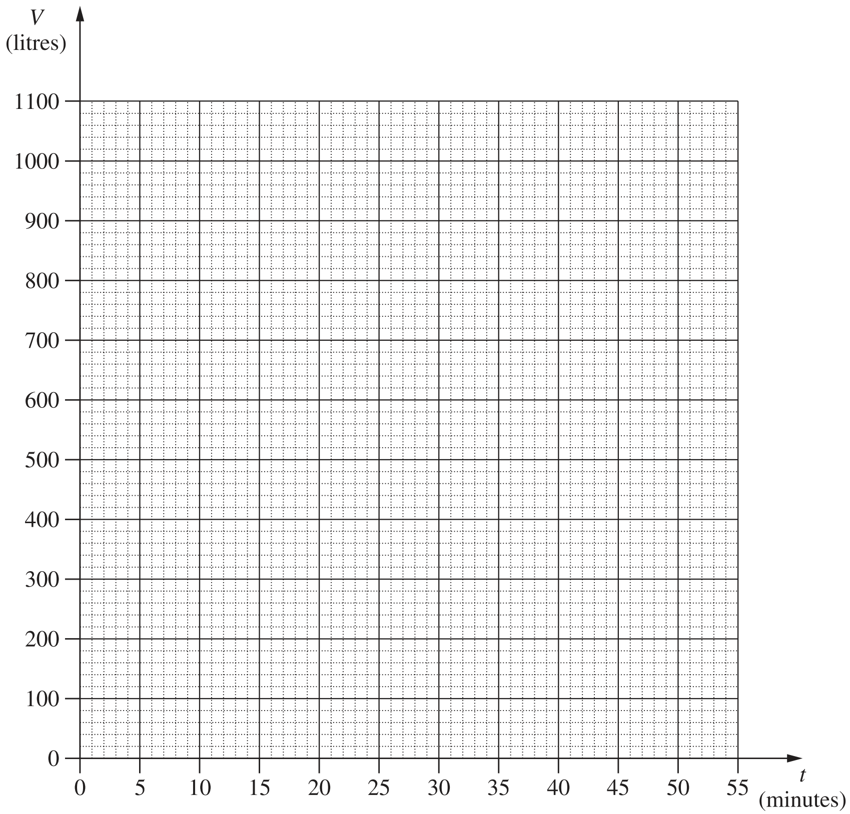
17. Algebra, STD2 A4 2020 HSC 24

There are two tanks on a property, Tank A and Tank B. Initially, Tank A holds 1000 litres of water and Tank B is empty.

a. Tank A begins to lose water at a constant rate of 20 litres per minute.

The volume of water in Tank A is modelled by $V = 1000 - 20t$ where V is the volume in litres and t is the time in minutes from when the tank begins to lose water.

On the grid below, draw the graph of this model and label it as Tank A. ***(1 mark)***



b. Tank B remains empty until $t = 15$ when water is added to it at a constant rate of 30 litres per minute. By drawing a line on the grid (above), or otherwise, find the value of t when the two tanks contain the same volume of water. ***(2 marks)***

c. Using the graphs drawn, or otherwise, find the value of t (where $t > 0$) when the total volume of water in the two tanks is 1000 litres. ***(1 mark)***

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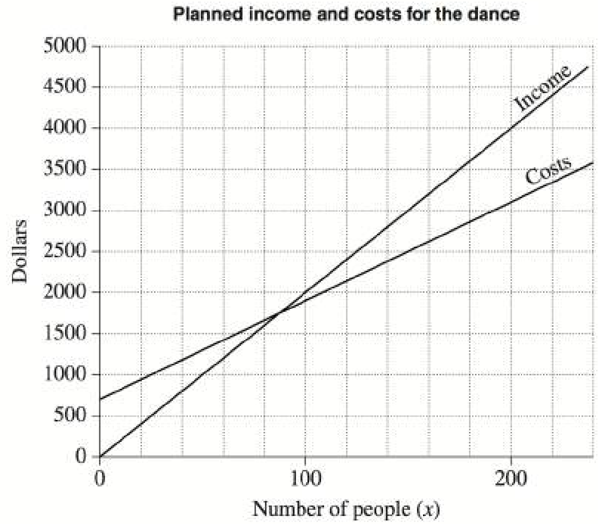
18. Algebra, STD2 A4 2005 HSC 28b

Sue and Mikey are planning a fund-raising dance. They can hire a hall for $400 and a band for $300. Refreshments will cost them $12 per person.

i. Write a formula for the cost ($C) of running the dance for x people. ***(1 mark)***

The graph shows planned income and costs when the ticket price is $20



ii. Estimate the minimum number of people needed at the dance to cover the costs. ***(1 mark)***

iii. How much profit will be made if 150 people attend the dance? ***(1 mark)***

Sue and Mikey plan to sell 200 tickets. They want to make a profit of $1500.

iv. What should be the price of a ticket, assuming all 200 tickets will be sold? ***(3 marks)***

19. Algebra, STD2 A4 SM-Bank 4

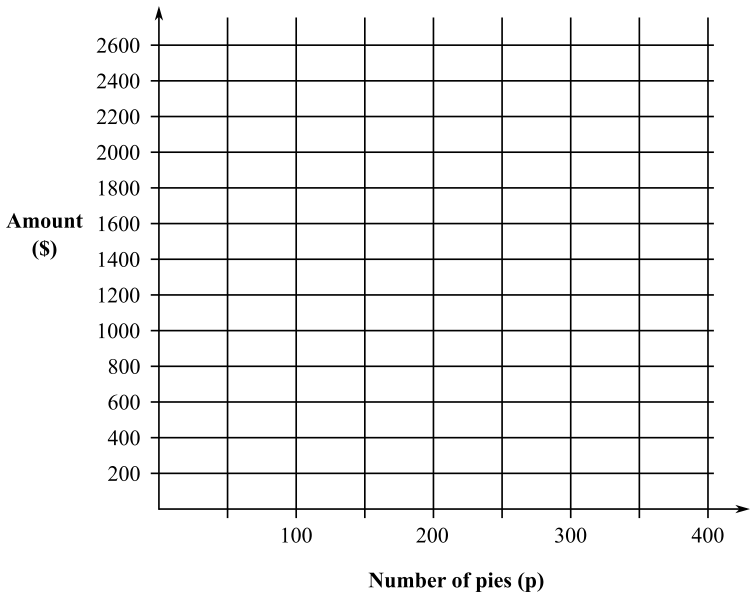
Penny is a baker and makes meat pies every day.

The cost of making p pies, $\$C$, can be calculated using the equation

$C = 675 + 3.5p$

Penny sells the pies for $5.75 each, and her income is calculated using the equation

$I = 5.75p$



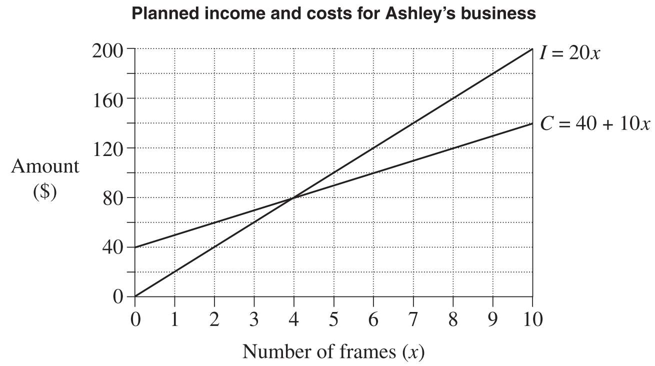
i. On the graph, draw the graphs of C and I. ***(2 marks)***

ii. On the graph, label the breakeven point and the loss zone. ***(2 marks)***

20. Algebra, STD2 A4 2010 HSC 24b

Ashley makes picture frames as part of her business. To calculate the cost, C, in dollars, of making x frames, she uses the equation $C = 40 + 10x$.

She sells the frames for $20 each and determines her income, I, in dollars, using the equation $I = 20x$.

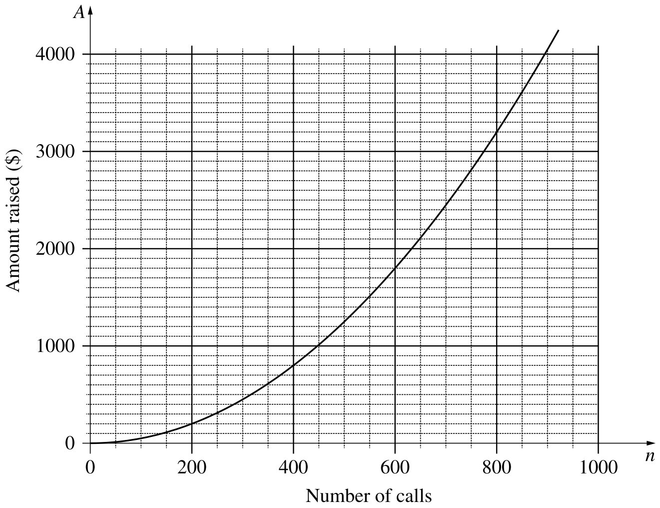


Use the graph to solve the two equations simultaneously for x and explain the significance of this solution for Ashley's business. ***(2 marks)***

21. Algebra, STD2 A4 2015 HSC 28f

A charity seeks to raise money by telephoning people at random from a call centre and asking them to donate.

Over the years, this charity has found that the amount of money raised $(\$A)$ is related to the number of telephone calls made (n). A graph of this relationship is shown.



It costs the charity $2100 per week to run the call centre. It also costs an average of 50 cents per telephone call.

i. Write an equation to represent the total cost, C, of running the call centre for a week in which n phone calls are made. ***(1 mark)***

ii. By graphing this equation on the axes above, determine the number of phone calls the charity needs to make in order to break even. ***(2 marks)***

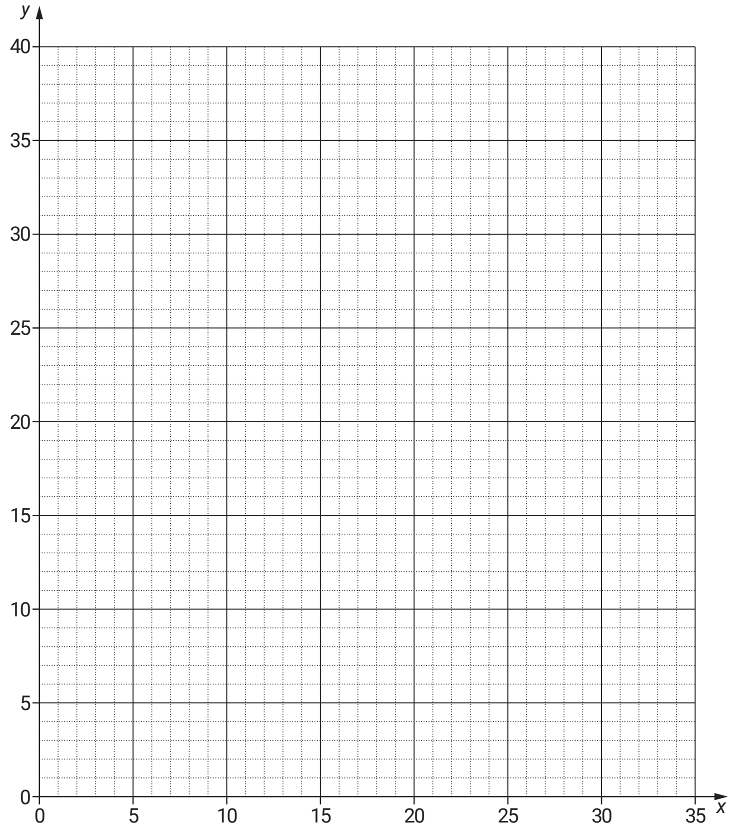
22. Algebra, STD2 A4 2021 HSC 34

In a park the only animals are goannas and emus. Let x be the number of goannas and let y be the number of emus.

The number of goannas plus the number of emus in the park is 31. Hence $x + y = 31$.

Each goanna has four legs and each emu has two legs. In total the emus and goannas have 76 legs.

By writing another relevant equation and graphing both equations on the grid on the following page, find the number of goannas and the number of emus in the park. ***(4 marks)***



Number of goannas = ______________

Number of emus = ________________

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Worked Solutions

1. Algebra, STD2 A4 SM-Bank 6 MC

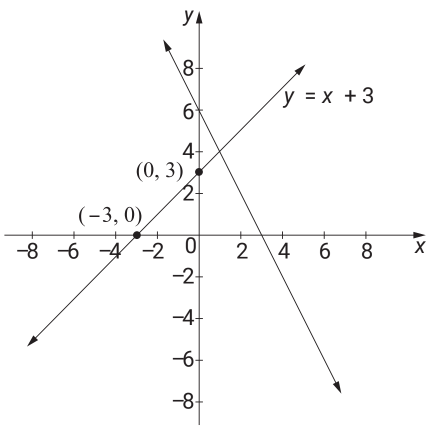
Solution occurs at the intersection of the two lines.

$\Rightarrow B$

2. Algebra, STD2 A4 2017 HSC 17 MC

Solution 1

From graph, intersection at (1,4)



$\Rightarrow B$

Solution 2

$y = 6 – 2x \quad \ldots (1)$

$y = x + 3 \quad \ldots (2)$

Substitute (2) into (1)

$x + 3 = 6 – 2x$

$3x = 3$

$x = 1$

When $x = 1, y = 4$

$\Rightarrow B$

3. Algebra, STD2 A4 SM-Bank 7 MC

Line 1: $x + y = 4$

Line 2: $x – y = 4$

Intersection at $(4, 0)$.

$\Rightarrow A$

Worked Solutions

4. Algebra, STD2 A4 2011 HSC 20 MC

When $x = 500,\ I = 100 \times 500 = \$50\ 000$

Breakeven when $x = 200$ (from graph)

When $x = 200,\ I = 100 \times 200 = \$20\ 000$

$\text{Difference} = 50\ 000 - 20\ 000$

$= \$30\ 000$

$\Rightarrow C$

♦ *Mean mark 50%*

COMMENT: Students can read the income levels directly off the graph to save time and then check with the equations given.

5. Algebra, STD2 A4 2004 HSC 16 MC

By elimination

$y = 2x – 5$ cuts the y-axis at -5

$\therefore$ Cannot be A or B

$y = x + 6$ cuts the y-axis at 6

AND has a positive gradient

$\therefore$ Cannot be C

$\Rightarrow D$

6. Algebra, STD2 A4 2024 HSC 14 MC

Cost increases by \$2.50 for each extra pizza.

Revenue increases by \$8 for each extra pizza.

Profit (additional) $= 8 - 2.5 = \$5.50$ per pizza

$\Rightarrow B$

7. Algebra, STD2 A4 2018 HSC 27d

i. Hourly rate $(A) = 360 \div 3$

$= \$120$

ii. $m =$ hourly rate

Find m, given $c = 360$, when $x = 3$ and $b = 75$

$360 = m \times 3 + 75$

$3m = 285$

$m = 95$

$\therefore c = 95x + 75$

iii. Cost $(A) = 120 \times 5 = \$600$

Cost $(B) = 95 \times 5 + 75 = \$550$

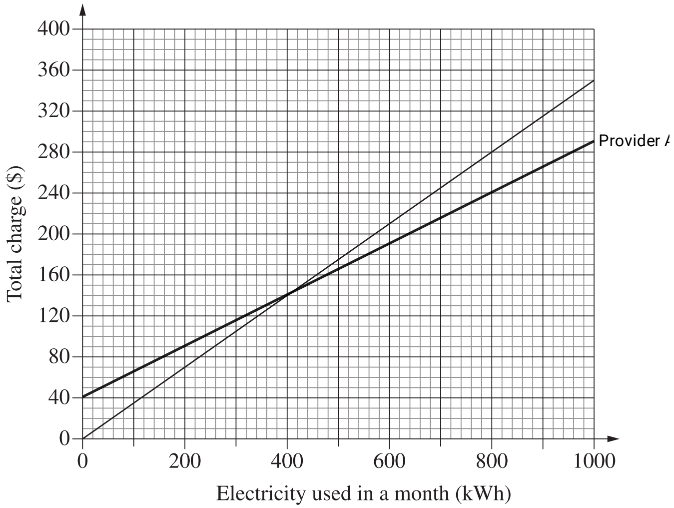
$\therefore$ Company B's hiring cost is \$50 cheaper.

8. Algebra, STD2 A4 2023 HSC 21

a.

Electricity used in a month (kWh)	0	400	1000
Monthly charge ($)	40	140	290

b.



c. $A_{\text{charge}} = B_{\text{charge}}$ at intersection.

$\therefore$ Same charge at 400 kWh

d. Cost at 800 kWh:

Provider A: $40 + 0.25 \times 800 = \$240$

Provider B: $0.35 \times 800 = \$280$

$\therefore$ Provider A is cheaper by \$40.

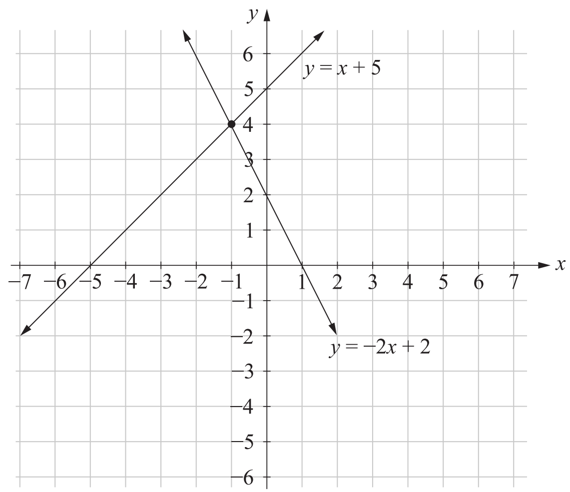
9. Algebra, STD2 A4 SM-Bank 5

Table of coordinates: $y = x + 5$

x	−5	−4	−2	0
y	0	1	3	5

Table of coordinates: $y + 2x = 2 \Rightarrow y = -2x + 2$

x	−2	−1	0	1
y	6	4	2	0



From graph (and table), intersection occurs at $(-1, 4)$.

10. Algebra, STD2 A4 SM-Bank 27

i. Fixed Costs $= 650 + 850$

$= \$1500$

Variable Costs $= \$25x$

$\therefore\ \$C = 1500 + 25x$

ii. From the graph

Costs $=$ Income when $x = 60$

(i.e. where graphs intersect)

iii. When $x = 80$:

Income $= 80 \times 50$

$= \$4000$

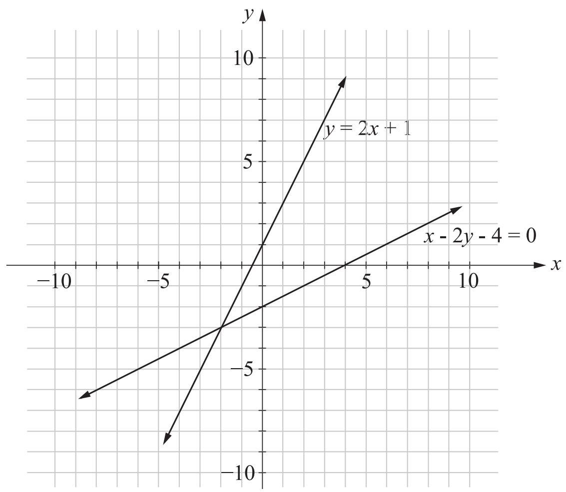
$\$C = 1500 + 25 \times 80$

$= \$3500$

$\therefore$ Profit $= 4000 – 3500$

$= \$500$

11. Algebra, STD2 A4 2014 HSC 26d



Solution is at the intersection: $x = -2,\ y = -3$

12. Algebra, STD2 A4 SM-Bank 6

Substitute $(5, 2)$ into $y = 2x – 8$

LHS $= y = 2$

RHS $= 2(5) – 8 = 2$

$\therefore$ LHS $=$ RHS

Substitute $(5, 2)$ into $x – 4y + 3 = 0$

LHS $= 5 – 4(2) + 3$

$= 0$

$=$ RHS

$\Rightarrow (5, 2)$ satisfies both equations.

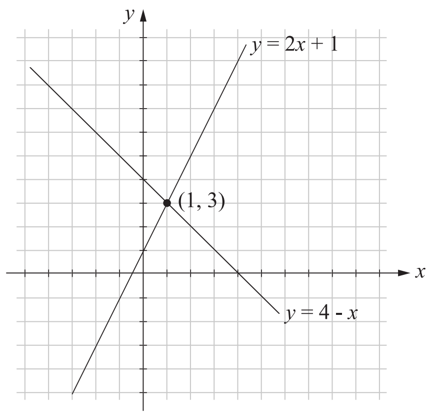
$\therefore$ Student is correct.

13. Algebra, STD2 A4 SM-Bank 7

Graphing $y = 2x + 1$:

y-intercept $= 1$

Gradient $= 2$



$\therefore$ Point of intersection is $(1, 3)$.

14. Algebra, STD2 A4 EQ-Bank 8

Graphing $4x + 3y = 20$

$$y = -\frac{4}{3}x + \frac{20}{3}$$

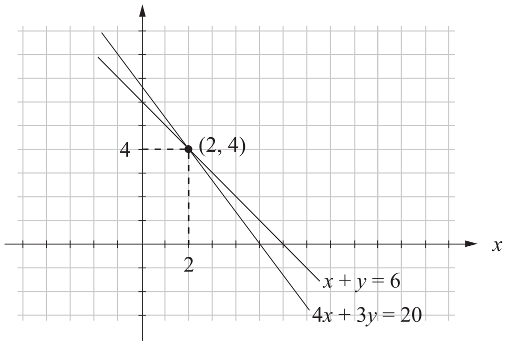
y-intercept $= \left(0, \frac{20}{3}\right)$

x-intercept $= (5, 0)$

Gradient $= -\frac{4}{3}$

Second equation:

$x + y = 6$



From the graph,

Sequoia sold 2 and Raven sold 4.

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15. Algebra, STD2 A4 SM-Bank 3

i. $F = \dfrac{9(-20)}{5} + 32$

$= -4°F$

ii. The two graphs intersect at a temperature where Celcius and Farenheit are the same.

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16. Algebra, STD2 A4 2019 HSC 36

a. 20 (x-value at intersection)

b. Find equations of both lines:

$(0, 500)$ and $(20, 800)$ lie on C

$$m_C = \frac{800 - 500}{20 - 0} = 15$$

$\Rightarrow C = 500 + 15x$

$(0, 0)$ and $(20, 800)$ lie on R

$$m_R = \frac{800 - 0}{20 - 0} = 40$$

$\Rightarrow R = 40x$

Profit $= R - C$

Find x when Profit = \$1900:

$$1900 = 40x - (500 + 15x)$$
$$25x = 2400$$
$$x = 96$$

♦♦ *Mean mark 28%.*

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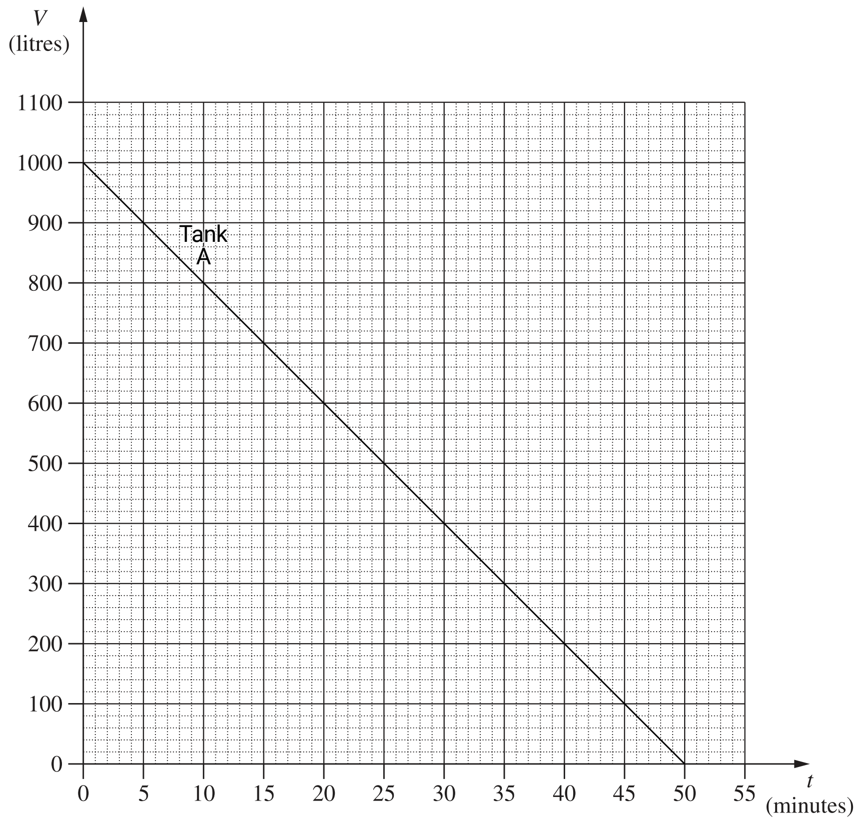
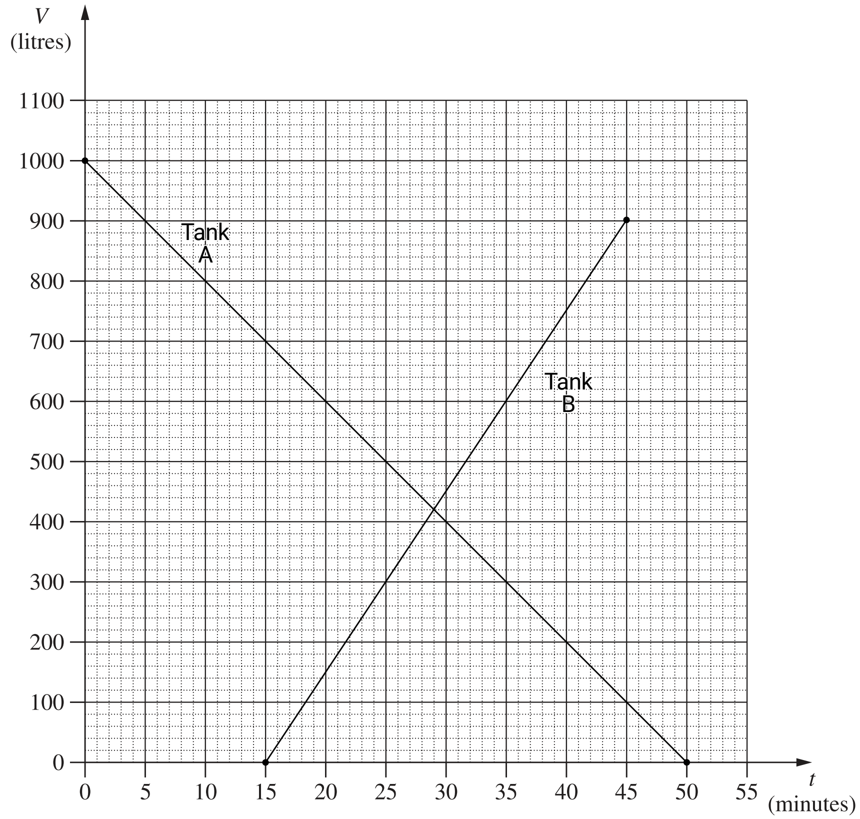
17. Algebra, STD2 A4 2020 HSC 24

a. Tank A will pass trough $(0, 1000)$ and $(50, 0)$



b. Tank B will pass through $(15, 0)$ and $(45, 900)$



By inspection, the two graphs intersect at $t = 29$ minutes

c. Strategy 1

By inspection of the graph, consider $t = 45$

Tank A $= 100$ L, Tank B $= 900$ L

$\therefore$ Total volume $= 1000$ L when $t = 45$

♦♦ *Mean mark part (c) 22%.*

Strategy 2

$$\text{Total Volume} = \text{Tank A} + \text{Tank B}$$

$$1000 = 1000 – 20t + (t – 15) \times 30$$

$$1000 = 1000 – 20t + 30t – 450$$

$$10t = 450$$

$$t = 45 \text{ minutes}$$

18. Algebra, STD2 A4 2005 HSC 28b

i. $\$C = 400 + 300 + (12 \times x)$

$= 700 + 12x$

ii. Using the graph intersection

Approximately 90 people are needed to cover the costs.

iii. If 150 people attend

$\text{Income} = 150 \times \$20$

$= \$3000$

$\text{Costs} = 700 + (12 \times 150)$

$= \$2500$

$\therefore\ \text{Profit} = 3000 - 2500$

$= \$500$

iv. Costs when $x = 200$:

$C = 700 + (12 \times 200)$

$= \$3100$

Income required to make \$1500 profit

$= 3100 + 1500$

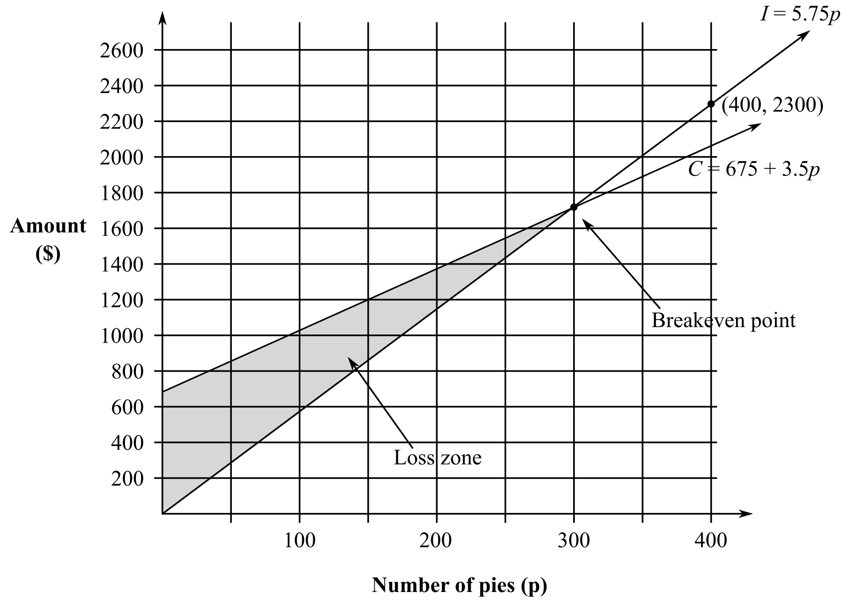
$= \$4600$

$\therefore\ \text{Price per ticket} = \dfrac{4600}{200}$

$= \$23$

19. Algebra, STD2 A4 SM-Bank 4

i.



ii. Loss zone occurs when $C > I$, which is shaded in the diagram above.

20. Algebra, STD2 A4 2010 HSC 24b

From the graph, intersection occurs at $x = 4$

$\Rightarrow$ Break-even point occurs at $x = 4$

i.e. when 4 frames sold

$\text{Income} = 20 \times 4 = \$80$ is equal to

$\text{Costs} = 40 + (10 \times 4) = \$80$

♦ *Mean mark 36%.*
MARKER'S COMMENT: The intersection on the graph is the same point at which the two simultaneous equations are solved for the given value of x.

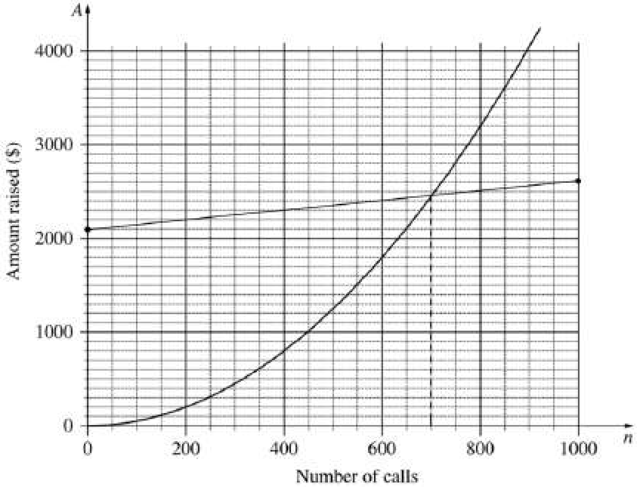
If < 4 frames sold $\Rightarrow$ LOSS for business
If > 4 frames sold $\Rightarrow$ PROFIT

21. Algebra, STD2 A4 2015 HSC 28f

i. $C = \$2100 + \$0.50n$

♦ *Mean marks of 48% and 32% for parts (i) and (ii) respectively.*

ii.



From the above graph, the charity needs to make 700 calls to break even.

22. Algebra, STD2 A4 2021 HSC 34

Total goanna legs $= 4x$

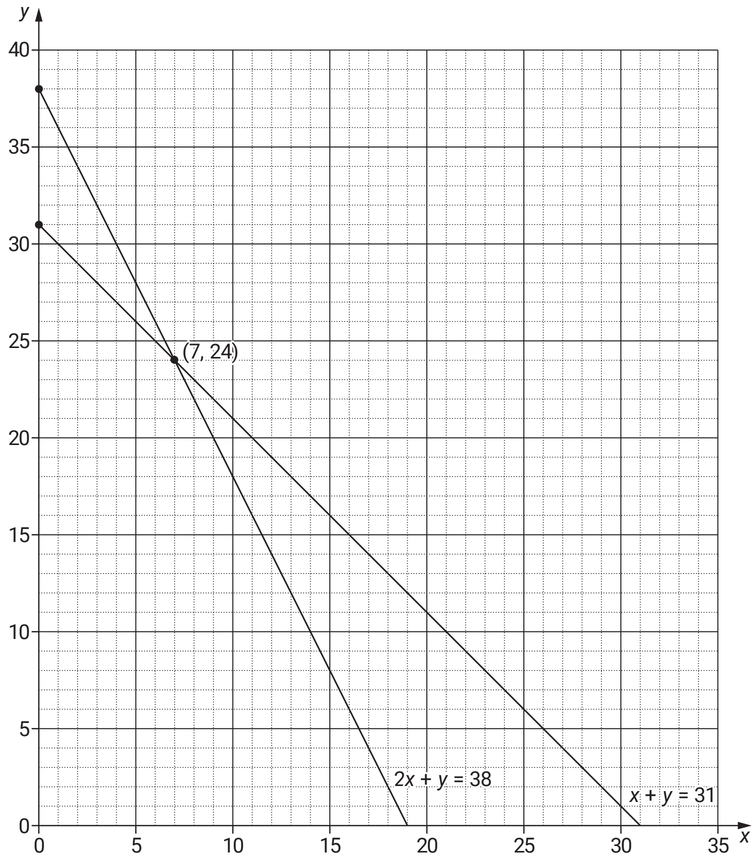
Total emu legs $= 2y$

♦♦♦ *Mean mark 29%.*

$$4x + 2y = 76$$

$$2x + y = 38 \ldots (1)$$

$$x + y = 31 \ldots (2)$$



Number of goannas $= 7$

Number of emus $= 24$

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